

EviTrace: Modeling Evidence-to-Claim Reasoning Traces for Fact Checking

Anonymous submission

Abstract

Large Language Models (LLMs) combined with Retrieval-Augmented Generation (RAG) have become a popular paradigm for automated fact-checking, yet most current systems treat evidence as an undifferentiated bag of texts to be consumed in one pass. This flat aggregation obscures the logical dependencies between individual pieces of evidence, making it difficult to trace how a conclusion is actually reached. Furthermore, the reasoning process itself often lacks a clear step-by-step audit trail, which undermines both interpretability and trustworthiness. To this end, we propose EviTrace, a framework that explicitly constructs structured evidence-to-claim reasoning traces by modeling the stepwise transition from each evidence snippet to intermediate and final verification judgments. Specifically, EviTrace takes as input an evidence pool that is re-chunked and organized and obtained through multi-path retrieval, together with the decomposed claims; it uses a relational graph between evidence units and Atom sub-claims as a medium, employs structural gain to select positive preference pairs for training a selector that simultaneously attends to multi-dimensional information, and finally relies on this selector and the state-change trajectory to generate an ordered evidence trajectory, which is then used for downstream verification. Experiments across benchmarks show that EviTrace consistently outperforms existing approaches, with notable gains in both accuracy and reasoning coherence. The results suggest that shifting the focus from evidence quantity to evidence organization offers a promising direction for building more interpretable and reliable fact-checking systems.

Code — ANONYMIZED_URL

Introduction

Automated fact checking has evolved from predicting veracity solely from claims or metadata to explicitly retrieving external evidence and jointly predicting a verdict. LIAR provides real-world political claims with fine-grained labels, but does not include gold evidence for retrieval (Wang 2017); FEVER combines evidence-sentence retrieval

with SUPPORTS/REFUTES/NOT-ENOUGH-INFO classification (Thorne et al. 2018); HoVer further extends verification to multi-hop reasoning across multiple documents (Jiang et al. 2020); and SciFact requires systems to identify evidence and determine stance jointly within a corpus of scientific abstracts (Wadden et al. 2020). In real-world settings, CofCED performs cascaded selection of reports and sentences from raw reports associated with a claim and introduces the LIAR-RAW and RAWFC datasets (Yang et al. 2022), whereas AVeriTeC extends the evidence source to the open Web (Schlichtkrull, Guo, and Vlachos 2023). The development of these tasks requires systems not only to “find relevant content,” but also to decide which evidence the final verifier should see.

Existing systems compress, graph-aggregate, or sequentially rank evidence before verification (Yang et al. 2022; Zhou et al. 2019; Aly and Vlachos 2022; Zhang and Gao 2024).

However, evidence organization is typically treated as latent computation or learned from downstream feedback, rather than exposed as an auditable source of supervision. Claim decomposition provides a natural index for evidence; we therefore ask: can claim-indexed evidence and its induced atom-state transitions be organized into an explicit, auditable trace that supports claim verification without gold evidence traces or verifier feedback?

We propose an atom-aware, structure-induced approach to evidence organization. Given a claim and its associated reports, the system sequentially performs claim atomization, claim-aware chunking, Atom-Union retrieval, and Evidence Map annotation. During training, structural gains at each rollout state determine a deterministic structure-induced ordering, and a nonnegative linear structural marginal score is learned from winner-vs-rest pairs. At inference time, the frozen selector accesses the full candidate pool and produces an audit ordering. The degree of evidence resolution and an evidence-capacity policy then determine the evidence prefix actually visible to the verifier, which is used to train the final verifier.

Our main contributions are as follows:

1. We introduce a structured evidence-to-claim trace representation that records selected evidence, typed candidate-atom relations, and stepwise atom-state transitions.
2. We induce prefix-dependent preferences from structural

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state gains and train a state-conditioned selector to construct these traces without gold evidence orders, verdict labels, or verifier feedback.

3. We evaluate the method on LIAR-RAW, RAWFC, and the cross-domain dataset SciFact, and conduct detailed mechanism analysis to identify the sources of gains; meanwhile, we conduct a double-blind evaluation experiment on LLM annotation results to ensure credibility.

Related Work

Fact Checking over Raw Reports

Raw-report fact checking operates on news reports that are relevant to a claim but have not yet been curated into a final explanation by human fact-checkers. CofCED first performs cascaded selection of explanatory sentences from the top- K reports and introduces the LIAR-RAW and RAWFC datasets (Yang et al. 2022). DeReC uses dense sentence retrieval with a compact DeBERTa classifier, emphasizing the efficiency and scalability of retrieval in this setting (Qazi, McCrae, and Nasir 2025). Methods from the LLM era further reshape the evidence-to-verdict interface: L-Defense generates competing supporting and opposing explanations (Wang et al. 2024); RAFTS makes verdicts through retrieval and contrastive argumentation (Yue et al. 2024); DelphiAgent reaches consensus through multi-role, multi-round feedback (Xiong et al. 2025); and G-Defense and KG-CRAFT introduce a subclaim graph and knowledge-graph-guided contrastive reasoning, respectively (Wang et al. 2026; Lourenço et al. 2026). FFRR trains a retrieval policy using fine-grained task feedback from a black-box LLM (Zhang and Gao 2024). Most of these methods employ complex forms of evidence organization, such as graph-based or explanation-based reasoning, while approaches based on verifier feedback inevitably introduce a black box that impedes auditing.

Claim Decomposition and Structured Evidence

Claim decomposition is commonly used to reduce the difficulty of retrieving and verifying evidence for complex claims. Literal/implied subquestion generation transforms complex claims into answerable subquestions (Chen et al. 2022); ProgramFC represents a complex verification process as a program composed of multiple subtasks (Pan et al. 2023); and wild-evidence verification combines claim decomposition, two-stage retrieval, and claim-focused summaries to process open corpora (Chen et al. 2024). Meanwhile, GEAR, KGAT, G-Defense, and KG-CRAFT have demonstrated the role of graph structures in evidence aggregation, subclaim dependency modeling, and contrastive reasoning (Zhou et al. 2019; Liu et al. 2020; Wang et al. 2026; Lourenço et al. 2026).

Sequential Evidence Ordering and Sufficiency

Prior work has studied iterative evidence selection, compact evidence sets, and sufficient prefixes. DQN-based selection formulates precise evidence search as a sequential decision problem (Wan et al. 2021); natural-logic-guided retrieval selects subsequent documents and determines when

to stop according to the proof state induced by retrieved evidence (Aly and Vlachos 2022); insufficient-evidence analysis shows that verifiers may not reliably recognize missing information (Atanasova et al. 2022). User-centric evidence ranking more directly studies complementary evidence and early sufficient prefixes, showing that incremental ranking and one-shot ranking have different capabilities (Alt et al. 2026).

Together, these studies show that the contribution of an evidence candidate depends on the currently selected context. Our distinction lies in the source of supervision: our selector does not depend on gold evidence, verdict-conditioned rewards, or other downstream feedback. Instead, it uses state transitions in a typed claim–evidence map over a fixed candidate pool as the primary signal, with map quality, retrieval relevance, and a frozen candidate index used for deterministic tie-breaking, thereby producing prefix-dependent structural selection preferences.

Methodology

Problem Definition

Given a claim c , an evidence collection $\mathcal{R}$, and a label space $\mathcal{Y}$, upstream processing constructs claim atoms $\mathcal{A}$, candidate evidence units $\mathcal{C} = \{u_j\}_{j=1}^n$, typed candidate–atom alignments M . We represent the resulting evidence-organization instance as

$$\mathcal{X} = (c, \mathcal{A}, \mathcal{C}, M).$$

The objective is to organize $\mathcal{C}$ into an evidence trace

$$\mathcal{T} = ((u_{\pi_1}, H^{(0)} \rightarrow H^{(1)}), \dots, (u_{\pi_t}, H^{(t-1)} \rightarrow H^{(t)})),$$

where the value of each candidate depends on the previously selected prefix. We use reasoning trace to denote an external, inspectable sequence of evidence selections and atom-state transitions, rather than the verifier’s latent chain-of-thought. Let Δ_{str} denote its structural marginal contribution. A desirable ordering is characterized by

$$J(\pi; \mathcal{X}) = \sum_{t=1}^n \gamma_t \Delta_{\text{str}}(u_{\pi_t} \mid \mathcal{T}_{<t}, \mathcal{X}), \gamma_1 \geq \dots \geq \gamma_n \geq 0,$$

which encourages structurally useful evidence to appear earlier, under the dynamic stopping strategy P , an order-preserving projection produces the verifier-visible prefix

$$\mathcal{T}^{\text{vis}} = P(\mathcal{T}),$$

from which the final prediction is obtained:

$$\hat{y} = \arg \max_{y \in \mathcal{Y}} p_{\theta_{\text{ver}}}(y \mid c, \mathcal{A}, \mathcal{T}^{\text{vis}}).$$

Overview. Our proposed EviTrace consists of three stages: (1) An Upstream Input Construction that decomposes the claim and constructs an Atom-Union candidate pool; (2) An Evidence Organization that builds a typed Evidence Map and learns a prefix-dependent selector from map-induced pairwise preferences; and (3) A Downstream Verification that passes an order-preserving prefix of the resulting trace to the downstream verifier. The overall method is illustrated in Figure 1.

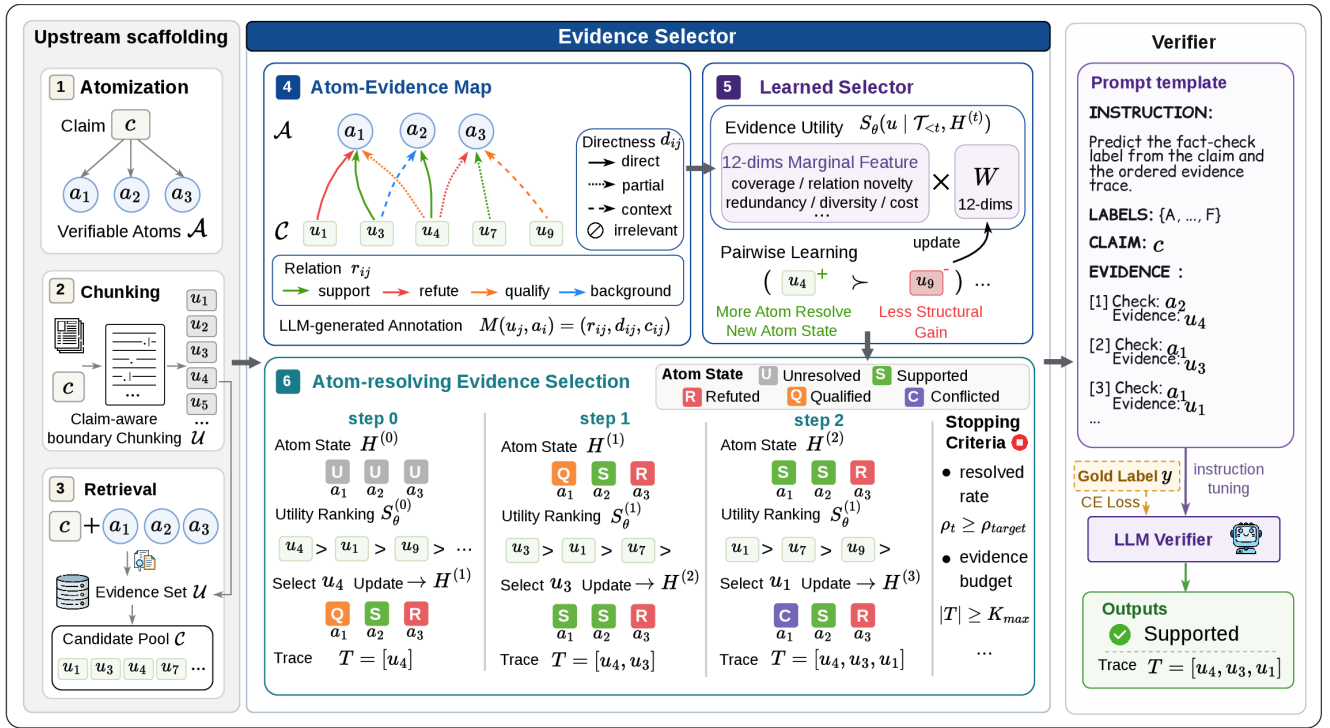


Figure 1: Overview of EviTrace. A typed Evidence Map induces prefix-conditioned structural preferences over Atom-Union candidates. A state-conditioned selector converts these preferences into an ordered, source-linked trace, whose budgeted prefix is passed to the verifier.

Upstream Input Construction

Claim Atomization. We decompose the claim into independent, semantically complete atoms $\mathcal{A} = \{a_i\}_{i=1}^m$ and corresponding retrieval queries. The atomizer accesses neither external reports nor fact labels and may not introduce information absent from the claim. We implement it with constrained LLM generation. Detailed prompts and implementation configurations are included in the supplementary material.

Claim-aware Evidence Chunking. We segment each report into multi-sentence evidence units $u_j = [s_a, \dots, s_b]$ to preserve local context while limiting irrelevant text. Boundaries are placed at local peaks of semantic or claim-relevance change:

$$b_i = w_{\text{sem}} [1 - \cos(\mathbf{h}_i, \mathbf{h}_{i+1})] + w_{\text{rel}} |r(s_i, c) - r(s_{i+1}, c)|, \\ i \in \mathcal{B}_{\text{split}} \iff b_i \geq \mu_b + \lambda \sigma_b \wedge b_i \geq \max(b_{i-1}, b_{i+1}).$$

Here, $\mathbf{h}_i$ and $r(s_i, c)$ denote the representation and claim relevance of sentence s_i , respectively; the selected local peaks form $\mathcal{B}_{\text{split}}$.

Atom-Union Candidate Pool Construction. To balance whole-claim relevance and atom coverage, we retrieve from the query views

$$\mathcal{Q} = \{c, q_1, \dots, q_m\}$$

using a shared multi-signal score:

$$s(x, u_j) = \beta_1 \text{norm}(\cos(e_x, e_{u_j})) \\ + \beta_2 \text{norm}(\text{LexF1}(x, u_j)) \\ + \beta_3 \text{norm}(\text{BM25}(x, u_j)).$$

Here, $\text{norm}(\cdot)$ denotes score normalization within the current retrieval route, and e_x and e_{u_j} are dense representations of the query and evidence unit, respectively. BM25 follows the classical probabilistic relevance framework (Robertson and Zaragoza 2009).

The claim route and atom routes return

$$\mathcal{B} = \text{TopK}_{k_b} \{u_j : s(c, u_j)\}, \\ \mathcal{R}_i = \text{TopK}_{k_a} \{u_j : s(q_i, u_j)\},$$

respectively. We fuse the atom routes using reciprocal rank fusion (Cormack, Clarke, and Buettcher 2009), then combine them with the claim route, deduplicate candidates, and apply maximal marginal relevance (Carbonell and Goldstein 1998):

$$\mathcal{C} = \text{MMR}(\text{Dedup}(\mathcal{B} \cup \text{RRF}(\mathcal{R}_1, \dots, \mathcal{R}_m))).$$

Evidence Organization

Typed Evidence Map. The Evidence Map represents each candidate-atom pair as

$$M(u_j, a_i) = (r_{ij}, d_{ij}, c_{ij}),$$

where

$r_{ij} \in \{\text{support, refute, qualify, background, irrelevant}\}$,
 d_{ij} denotes ordinal directness, and $c_{ij} \in [0, 1]$ denotes annotation confidence. A constrained LLM annotator produces these fields; their human alignment is evaluated in the Experiments section.

For a prefix $\mathcal{T}_{<t}$, the map induces an atom state $h_i^{(t)} \in \{U, S, R, Q, C\}$, denoting uncovered, supported, refuted, qualified, or conflicting evidence, respectively. The complete prefix state is $H^{(t)} = \{h_i^{(t)}\}_{i=1}^m$.

Structure-Induced Preferences. At prefix state t , a deterministic lexicographic rule selects the highest-priority candidate u_t^+ , primarily according to its map-state gain, and pairs it with every remaining candidate:

$$\mathcal{P}_t = \{(u_t^+, u^-) : u^- \in \mathcal{C}_t \setminus \{u_t^+\}\}.$$

After appending the winner, we update $H^{(t)}$ and regenerate the pairs, making the preferences prefix-dependent. This supervision uses neither verdict labels, gold or teacher-provided evidence order, nor verifier feedback.

State-Conditioned Selector. We distill these discrete preferences into a continuous state-conditioned scorer, allowing structural coverage, relation novelty, redundancy, source diversity, annotation quality, retrieval relevance, and context cost to be traded off jointly.

Each candidate is represented by a 12-dimensional feature vector $\phi^{(t)}(u)$ covering state progress, atom and relation coverage, source and textual novelty, annotation quality, retrieval relevance, and context cost.

$$S_{\theta}^{\text{str}}(u \mid \mathcal{T}_{<t}, H^{(t)}) = b + \sum_{\ell \neq \text{cost}} w_{\ell} \phi_{\ell}^{(t)}(u) - w_c \phi_{\text{cost}}^{(t)}(u).$$

We parameterize all weights with softplus, preserving positive directions for benefit features and a negative direction for cost. The scorer is trained on $\mathcal{P} = \bigcup_t \mathcal{P}_t$ using pairwise logistic loss (Burges et al. 2005):

$$\mathcal{L}(\theta) = \frac{1}{|\mathcal{P}|} \sum_{(u^+, u^-) \in \mathcal{P}} \log(1 + \exp[-S_{\theta}^{\text{str}}(u^+) + S_{\theta}^{\text{str}}(u^-)]).$$

Evidence Ordering. At inference, the frozen scorer greedily selects

$$u_t = \arg \max_{u \in \mathcal{C}_t} S_{\theta}^{\text{str}}(u \mid \mathcal{T}_{<t}, H^{(t)}),$$

and recomputes the prefix-dependent state and features after each selection. Let

$$\rho_t = \frac{1}{|\mathcal{A}|} \sum_{i=1}^{|\mathcal{A}|} \mathbb{I}[h_i^{(t)} \neq U]$$

denote the resolved-atom rate. The trace stops at the first $t \geq k_{\min}$ satisfying $\rho_t \geq \rho_{\text{target}}$; otherwise, it terminates at $k_{\max}$ or candidate exhaustion.

Downstream Verification

The verifier prompt contains the claim, short atom cues, and the original evidence text from an order-preserving visible prefix $\mathcal{T}^{\text{vis}} = P_B(\mathcal{T})$, truncated only at evidence-unit boundaries. Labels are mapped to single-token answer choices through $g : \mathcal{Y} \rightarrow \{A, B, \dots\}$, and the verifier is trained by

$$\mathcal{L}_{\text{ver}} = - \sum_n \log p_{\theta_{\text{ver}}}(g(y_n) \mid \text{Render}(c_n, \mathcal{A}_n, \mathcal{T}_n^{\text{vis}})).$$

We fine-tune an autoregressive verifier with LoRA (Hu et al. 2022); backbone, rendering, and training details are reported in the Experiments section.

Experiments

To evaluate the effectiveness of our proposed EviTrace framework, we conduct extensive experiments to address the following research questions (RQ):

RQ1: How does the method perform compared to other fact-checking methods and across datasets from different domains?

RQ2 How do candidate construction, learned structured scoring, evidence order, and Evidence Map fields affect the performance of the method?

RQ3 What are the performance-cost curves of different evidence selection strategies?

RQ4: How reliable are the LLM-generated structural annotations?

Datasets and Evaluation Protocol

LIAR-RAW (Yang et al. 2022) contains 12,590 real-world political claims with six fine-grained labels and associated raw reports. **RAWFC**, introduced in the same work, contains 2,012 Snopes claims with three labels (*false*, *half*, and *true*), split into 1,612/200/200 training, validation, and test examples. We use the official splits.

SciFact (Wadden et al. 2020) contains 1.4K expert-written claims, 5,183 scientific abstracts, abstract-level stance annotations, and sentence-level rationale annotations. Because the complete official test set has not been publicly released, we report development-set results using the official micro-F1 scorer.

For cross-paper comparisons, we report published macro-averaged precision, recall, and F1-score. In our controlled ablation and capacity analyses, we additionally report accuracy computed from model predictions.

Implementation Details

LLM-based structural annotation. Both atom decomposition and Evidence Map structural annotation are performed through a unified LLM API using `deepseek-v4-flash` (DeepSeek-AI 2026). We disable thinking mode, set **temperature** to 0 and **top-p** to 1.0, and limit the maximum output to 2,048 tokens. We freeze all annotation artifacts and provide them in full in an anonymous repository for strict reproducibility.

Method / variant	Evidence access	LIAR-RAW			RAWFC		
		P	R	F1	P	R	F1
<i>Target-claim paired reports</i>							
dFEND [†] (?)	Paired reports	23.09	18.56	17.51	44.93	43.26	44.07
SBERT-FC [†] (?)	Paired reports	24.09	22.07	22.19	51.06	45.92	45.51
GenFE [†] (?)	Paired reports	28.01	26.16	26.49	44.29	44.74	44.43
GenFE-MT [†] (?)	Paired reports	18.55	19.90	15.15	45.64	45.27	45.08
MUSER [‡] (?)	Paired reports	27.80	26.58	26.52	47.47	47.50	47.48
CofCED (Yang et al. 2022)	Paired reports	29.48	29.55	28.93	52.99	50.99	51.07
L-Defense (GPT-3.5) (Wang et al. 2024)	Paired reports	30.55	32.20	30.53	61.72	61.01	61.20
L-Defense (LLaMA2) (Wang et al. 2024)	Paired reports	31.63	31.71	31.40	60.95	60.00	60.12
G-Defense (GPT-3.5) (Wang et al. 2026)	Paired reports	33.09	31.55	31.55	64.99	64.52	64.31
G-Defense (background) (Wang et al. 2026)	Paired reports	31.50	31.28	30.89	66.29	65.49	65.50
G-Defense (LLaMA-3.1-8B) (Wang et al. 2026)	Paired reports	34.17	32.37	32.49	60.97	61.23	60.76
DelphiAgent (GPT-4o) (Xiong et al. 2025)	Paired reports	31.33	28.36	28.36	68.05	<u>68.03</u>	<u>68.04</u>
<i>Broader or privileged evidence access</i>							
DeReC (Qwen) (Qazi, McCrae, and Nasir 2025)	Benchmark-corpus retrieval	35.94	32.24	33.13	65.58	64.56	64.60
DeReC (Nomic) (Qazi, McCrae, and Nasir 2025)	Benchmark-corpus retrieval	33.19	31.50	31.79	64.48	65.57	64.61
FFRR (d+q; GPT-3.5) (Zhang and Gao 2024)	Pooled benchmark corpus	34.50	32.60	33.50	56.50	57.40	57.00
ReAct (GPT-3.5 + Google) [§] (?)	Open-Web retrieval	33.20	29.00	31.00	51.20	48.50	49.80
Verify-and-Edit (GPT-3.5 + Google) [§] (?)	Open-Web retrieval	32.70	28.10	30.20	53.10	52.80	52.90
RAV (ϕ -4) (Shukla, Dutta, and Bhattacharyya 2025)	Gold evidence	–	–	29.33	–	–	67.53
EviTrace (LLaMA-3.1-8B; Ours)	Paired reports	<u>37.71</u>	<u>33.16</u>	<u>34.31</u>	<u>69.09</u>	69.06	69.00
EviTrace (Ministral-3-8B; Ours)	Paired reports	38.37	35.87	36.66	71.77	66.98	67.64

Table 1: Published LIAR-RAW and RAWFC test results (P/R/F1, %), grouped by evaluation-time evidence access. Values follow the evaluation implementations reported in the cited papers. Bold and underlined values denote the best and second-best results, respectively, among systems restricted to reports paired with the target claim. Benchmark-corpus, open-Web, and gold-evidence results are shown for context and are excluded from within-group ranking. [†]Raw-report adaptations and corresponding results are taken from Yang et al. (2022). [‡]The RAWFC/LIAR-RAW reproduction and results are reported by Wang et al. (2026). [§]The GPT-3.5+Google implementations and results are reported by Zhang and Gao (2024).

Retrieval and selector. For LIAR-RAW and RAWFC, we use BAAI/bge-base-en-v1.5 (Xiao et al. 2023) as the dense retrieval encoder. SciFact uses the same dense encoder for open-corpus retrieval. Preference training performs a 5-step rollout over the top-20 candidates for each example and runs for 30 epochs with a learning rate of 0.05, starting from an equal-weight neutral initialization. At inference time, audit ordering instead accesses the full candidate pool after deduplication and validity checks.

Verifier and context policy. On LIAR-RAW and RAWFC, we evaluate two 8B verifier backbones: Ministral-3-8B-Instruct-2512 (Mistral AI 2025) and Meta-Llama-3.1-8B-Instruct (?). SciFact and the single-verifier controlled ablation and capacity analyses use Ministral-3-8B-Instruct-2512. RQ2 additionally reports a balanced cross-verifier mechanism check with Qwen3-4B-Instruct-2507 and Meta-Llama-3.1-8B-Instruct. Both main-study verifiers are fine-tuned with LoRA and make deterministic decisions from the logits of candidate label tokens. Training uses 4-GPU ZeRO-2, BF16, a per-device batch size of 1, gradient accumulation of 4, a learning rate of 2×10^{-5} , evaluation and checkpoint saving every 100 steps, patience of 12, AdamW, weight decay of 0.01, a warmup ratio of 0.03,

a cosine-with-restarts scheduler, and LoRA with $r = 16$, $\alpha = 32$, dropout of 0.1, applied to the attention and MLP projections.

Overall Performance (RQ1)

This section evaluates EviTrace on LIAR-RAW, RAWFC, and SciFact. We first compare verification performance under explicitly separated evidence-access protocols and then report full-pipeline SciFact results.

Main comparison methods. We group the baselines by evaluation-time evidence access.

Among paired-report methods: **dFEND** (?): Couples veracity prediction with salient news-sentence and user-comment selection via co-attention. **SBERT-FC** (?): Ranks sentences by their Sentence-BERT similarity to the claim. **GenFE/GenFE-MT** (?): Perform DistilBERT-based veracity prediction separately or jointly with extractive explanation selection. **MUSER** (?) : Performs adaptive multi-step evidence retrieval and claim-evidence reasoning. **L-Defense** (Wang et al. 2024): Generates competing supporting and refuting justifications before defense-based inference. **G-Defense** (Wang et al. 2026): Extends L-Defense’s strategy with a dependency graph over decomposed subclaims; its

Method	Abstract Label-only	Abstract Label+Rationale
VERISCI [†] (Wadden et al. 2020)	52.10	50.00
Paragraph-Joint (FEVER-pretrained) (Li, Burns, and Peng 2021)	65.10	59.90
PrunE (Fang, Fu, and Torvik 2025)	63.21	60.10
QMUL-SDS (BioBERT-L) (Zeng and Zubiaga 2021)	64.52	61.29
VerT5erini (T5) (Pradeep et al. 2021)	65.07	61.72
ARSSJoint (BioBERT) (Zhang et al. 2021)	<u>66.70</u>	62.40
RerrFact (Rana et al. 2022)	64.59	<u>64.02</u>
EviTrace (Ministral-3-8B; Ours)	71.98	64.78

Table 2: Cross-paper SciFact development-set results under the official micro-F1 scorer (%). Published baselines follow their originally reported retrieval and candidate-construction protocols. Label+Rationale additionally requires a predicted rationale containing a gold rationale. [†]The VERISCI scores follow the development-set evaluation reported by Li, Burns, and Peng (2021); Wadden et al. (2020) is cited as the original method source.

variants add background context. **DelphiAgent** (Xiong et al. 2025): uses multiple LLM agents and iterative feedback to reach a consensus verdict.

Among broader-access systems,: **DeReC** (Qazi, McCrae, and Nasir 2025): Combines dense sentence retrieval with a DeBERTa classifier, using either Qwen or Nomic embeddings. **FFRR** (Zhang and Gao 2024): Optimizes document- and question-level retrieval using fine-grained feedback from GPT-3.5. **ReAct** (?): Interleaves reasoning traces with external-search actions. **Verify-and-Edit** (?): Retrieves external knowledge to revise low-consistency reasoning chains. **RAV** (Shukla, Dutta, and Bhattacharyya 2025): Iteratively generates questions, answers them, and predicts a label; its reported ϕ -4 configuration uses gold evidence.

SciFact baselines. **VERISCI** (Wadden et al. 2020): Pipelines TF-IDF retrieval, rationale selection, and stance classification. **Paragraph-Joint** (Li, Burns, and Peng 2021): Jointly predicts rationales and stance from paragraph representations. **ARSSJoint** (Zhang et al. 2021): Jointly optimizes retrieval, rationale selection, and stance prediction with rationale regularization. **QMUL-SDS** (Zeng and Zubiaga 2021): Uses BioBERT evidence filtering and two-step stance decisions; **VerT5erini** (Pradeep et al. 2021): Applies T5 to reranking, sentence selection, and labeling. **RerrFact** (Rana et al. 2022): Combines reduced abstract representations with loosely coupled evidence selection and two-stage classification. **PrunE** (Fang, Fu, and Torvik 2025): Learns to prune a sentence-level evidence graph.

Table 1 reports EviTrace with two verifier backbones. With Ministral-3-8B, EviTrace obtains F1 scores of 36.66 and 67.64 on LIAR-RAW and RAWFC, respectively; with

Variant	Pool	Scorer	Ord.	Acc.	F1
No evidence	–	–	–	30.70	31.02
Random	Raw	–	Rand.	31.25	31.68
Claim route only	Claim	Ret.	Score	33.81	34.51
Atom routes only	Atom	Ret.	Score	31.15	32.11
Retrieval only	Union	Ret.	Score	35.01	35.37
Fixed utility	Union	Fixed	Util.	34.85	34.92
Shuffled trace	Union	Learned	Rand.	31.70	32.90
Full	Union	Learned	Util.	35.97	36.66

Table 3: Selector-mechanism ablation on LIAR-RAW using the Ministral-3-8B verifier. Pool denotes the candidate-evidence pool; Union denotes the union of atom-routed candidates. Ret., Ord., Util., Rand., and F1 denote retrieval, order, utility, random, and Macro-F1, respectively.

LLaMA-3.1-8B, it obtains 34.31 and 69.00. Relative to the strongest prior systems under the target-claim paired-report setting, the best EviTrace variant improves F1 by 4.17 points on LIAR-RAW and 0.96 points on RAWFC. Compared specifically with the reported G-Defense LLaMA-3.1-8B variant, the EviTrace LLaMA variant improves F1 by 1.82 points on LIAR-RAW and 8.24 points on RAWFC. The table separately reports systems that retrieve from a benchmark-level corpus, query the open Web, or receive gold evidence. These rows characterize a broader empirical landscape but are not treated as controlled competitors because their evidence spaces differ from EviTrace’s target-claim paired-report setting.

KG-CRAFT (Lourengo et al. 2026) is discussed in the related work section, but it is not included in Table 1. Although its paper reported results on LIAR-RAW and RAWFC, it did not specify how its metrics were calculated, so we excluded these values from the numerical comparison and ranking.

Table 2 shows strong SciFact results on both abstract-level metrics. However, because the complete test split was unavailable, the set of comparable methods is limited, and the persuasiveness of these metrics should be strengthened by subsequently evaluating EviTrace on the test split.

Ablation Study (RQ2)

We ablate three stages of the selection pipeline: candidate construction, selector scoring, and trace ordering. We separately replace each Evidence Map field by a constant and retrain the selector over the resulting feature set. Unless otherwise stated, all controlled analyses in RQ2 and RQ3 use the Ministral-3-8B verifier; the balanced cross-verifier check described below is the exception.

Balanced cross-verifier mechanism check. To test whether the central selector contrasts in Table 3 transfer beyond a single verifier, we conduct a paired LIAR-RAW follow-up using Qwen3-4B-Instruct-2507 and Meta-Llama-3.1-8B-Instruct. Each backbone is fine-tuned under two complementary mixed-arm assignments and three seeds, giving 12 runs. Every training claim appears with exactly one evidence arm; the complementary assignment reverses that arm for every claim. Each

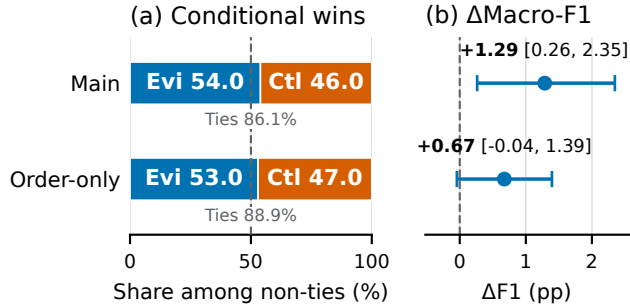


Figure 2: Balanced cross-verifier mechanism evaluation on LIAR-RAW. **(a)** Conditional paired correctness wins, normalized over predictions for which only one arm is correct. Evi and Ctl denote EviTTrace and the matched control; the corresponding tie rates (both correct or both incorrect) are 86.07% for Main and 88.87% for Order-only. *Main* compares the complete EviTTrace sequence with the source-score retrieval-only control at the same claim-specific evidence count. *Order-only* holds the evidence set and text fixed and compares EviTTrace order with source-score reordering. **(b)** Panel-level EviTTrace-minus-control Macro-F1 differences with gold-label-stratified, claim-clustered bootstrap 95% confidence intervals. Results aggregate two verifier backbones, two complementary training assignments, and three seeds; claims, rather than the 12 repeated run outputs, are the independent units. Holm-adjusted p -values are .029 for Main and .065 for Order-only.

assignment therefore contains 5,025 EviTTrace and 5,025 retrieval-ordered examples, balanced within gold-label $\times$ single-/multi-atom strata. Checkpoints are selected by the mean validation Macro-F1 over the two arms. We use the native chat template of each backbone and the same method-blind LIAR-6 prompt. Training uses LoRA with $r = 16$, $\alpha = 32$, dropout 0.05, attention and MLP projection targets, BF16 ZeRO-2, an effective batch size of 16, a learning rate of 2×10^{-5} , 12 epochs, and a maximum length of 2,048 tokens.

The *Main* comparison (1,250 claims) evaluates the final verifier-visible EviTTrace sequence against the source-score retrieval-only top- K_i sequence from the same Atom-Union pool, using the same final evidence count K_i but not enforcing equal token counts. The *Order-only* comparison fixes the EviTTrace evidence set and text and changes only its order: EviTTrace order versus source-score reordering. We exclude 98 claims whose two orders are identical, leaving 1,152 claims. Results are averaged first over assignment and seed, then equally over the two backbones. We compute gold-label-stratified, claim-clustered bootstrap intervals with 10,000 replicates and claim-level arm-swap tests with 100,000 randomizations; the two primary Macro-F1 tests receive Holm correction.

Table 3 shows that the claim and atom routes are complementary. Learned scoring improves over fixed weights, while shuffling the learned trace reduces Macro-F1 by 3.76 points. Table 4 shows that removing the full map reduces Macro-F1

Variant	Rel. (r)	Dir. (d)	Conf. (c)	Acc.	F1	$\bar{K}$
No relation	Backgr.	Orig.	Orig.	35.25	35.53	9.69
No directness	Orig.	Neutral	Orig.	34.37	34.16	9.69
Constant confidence	Orig.	Orig.	1.0	33.01	33.19	6.23
No map	Off	Off	Off	33.57	34.32	9.69
Full	Orig.	Orig.	Orig.	35.97	36.66	6.23

Table 4: Evidence-map component ablation on LIAR-RAW using the Ministral-3-8B verifier. Rel., Dir., and Conf. denote relation, directness, and confidence; F1 denotes Macro-F1.

by 2.34 points. Relation removal retains relatively strong verification performance but increases the average visible prefix from 6.23 to 9.69 units, whereas constant confidence causes the largest performance drop.

Figure 2 provides a cross-verifier check of the two central selector contrasts. In Main, EviTTrace obtains 33.49 Macro-F1 versus 32.20 for retrieval only, a gain of 1.29 points (95% CI [0.26, 2.35]; Holm-adjusted $p = .029$), with positive backbone-specific differences for both Qwen3 (+1.80) and LLaMA-3.1 (+0.77). Its conditional win rate among correctness-discordant pairs is 53.95%. This corroborates the Full-versus-Retrieval-only contrast and supports a small cross-verifier benefit from evidence selection and overall organization, although the interval still includes effects below the prespecified one-point SESOI. Under Order-only, EviTTrace obtains 33.27 versus 32.60 Macro-F1 ($\Delta = +0.67$; 95% CI [-0.04, 1.39]; Holm-adjusted $p = .065$) and a 52.97% conditional win rate. Both backbones have positive point estimates, but the pooled interval crosses zero. This same-set source-score reordering is a stricter order control, not the random *Shuffled trace* in Table 3; the two results are complementary. Together they suggest that trace organization is useful, but the cross-verifier experiment alone does not establish a reliable ordering-only improvement.

Evidence Capacity Sensitivity (RQ3)

We compare three capacity policies over a shared evidence ordering: fixed- k uses a fixed evidence count, $\text{minmax}(k_{\min}, k_{\max})$ allows structure-conditioned stopping within a bounded range, and budget- B limits the verifier-visible context by tokens.

Figure 3 shows that increasing fixed capacity continuously raises context cost without yielding consistent Macro-F1 improvements. In contrast, $\text{minmax}(5, 10)$ forms the strongest observed operating point: compared with fixed $k=9$, it improves Macro-F1 by 1.85 points while using 189.75 fewer prompt tokens on average. Meanwhile, a larger token-level budget strategy likewise does not translate into more cost-effective performance, which indicates that the dynamic selection strategy starting from 5 evidences is superior to the fixed strategy.

Human Audit (RQ4)

We audit claim atomization on 250 atoms from 200 claims and Evidence Map annotation on 250 candidate-atom pairs

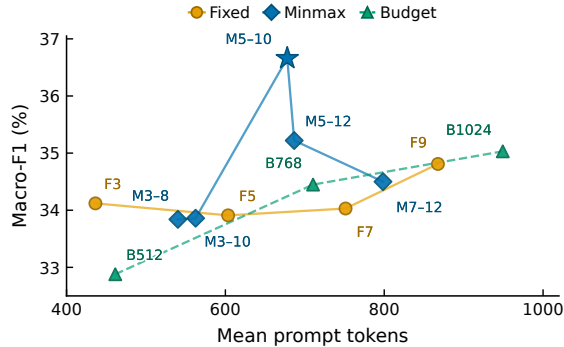


Figure 3: Evidence-capacity sensitivity on LIAR-RAW using the Ministral-3-8B verifier. Points show Macro-F1 against the mean prompt tokens presented to the verifier; the upper-left direction is preferable. Colors, markers, and lines distinguish fixed- k , minmax, and budget- B policies. The star marks the best observed configuration, minmax(5, 10).

from 200 events. Atomization disagreements are resolved by a blinded third annotator, whereas both Evidence Map annotations are retained as empirical human references.

Table 5 summarizes inter-annotator agreement and model-human alignment. Atomization is highly stable, while Evidence Map labels remain useful but uncertainty-sensitive, particularly for Relation and confidence calibration.

Case Study

Tables 3–5 characterize the complementary preconditions of auditable trajectories. The ablation experiments show that the evidence order and the typed mapping fields have practical effects at the behavioral level, while human auditing delimits the semantic reliability of the annotations presented in the trajectories. To illustrate how these quantities jointly explain an individual selection trajectory, we present an example from LIAR-RAW in Figure 4.

The example decomposes the given claim into two verifiable atoms, and the selected evidence units give rise to distinct state transitions. Subsequent selections add qualifying context and complementary coverage, ultimately forming a compact trajectory that contains both decisive evidence and contextual evidence.

Notably, candidate evidence that is highly relevant but unable to resolve the atoms, such as E04 and E07, was not included in the final trajectory. This example demonstrates that the selector exploits the mapping-induced structural gain rather than relying solely on retrieval scores, thereby producing an ordered evidence path.

Conclusion

We propose **EviTrace**, a framework guided by atom-state transitions that leverages evidence organization as a supervision signal for fact-checking. EviTrace takes an evidence candidate pool as input, uses structural gain as a proxy to construct a preference-learning-based selector, and organizes

(a) Claim atomization: 250 atoms from 200 claims

Criterion	Unit	Exact (%)	AC1	Adj. pass (%)
Faithfulness	Atom	95.72	.955	99.22
Atomicity	Atom	88.72	.866	95.72
Coverage	Claim	95.48	.953	99.00
Strict joint pass	Claim	–	–	93.50

(b) Evidence Map: 250 candidate-atom pairs from 200 events

Field	Human reference	LLM result	Additional diagnostic
Relation	Exact 51.6%; $\kappa = .419$	Exp. 46.8%; Any 60.4%	–
Directness	Exact 65.6%; $\rho = .817$	Exp. 56.2%	Within-one 93.6%
Confidence	Human mean .872/.841	LLM mean .561	ECE .392; Brier .329

Table 5: Human audit of the LLM-generated structural annotations. Exp. denotes the expected match to an annotator, and Any denotes a match to either observed human annotation.

evidence according to state-transition prefixes, thereby generating ordered trajectories for downstream verification.

On two 8B verifier backbones, EviTrace remains consistently competitive, although the choice of the optimal verifier is dataset-dependent: Ministral performs better on LIAR-RAW, while LLaMA performs better on RAWFC. Single-verifier controlled analyses establish the contributions of candidate construction, structured scoring, and evidence ordering under the Ministral configuration, while the balanced Qwen3/LLaMA follow-up finds a small cross-verifier gain for evidence selection and overall organization but inconclusive ordering-only evidence. Meanwhile, human evaluation shows that claim atomization is stable, and that Evidence Map annotations, although still sensitive to uncertainty, retain practical value. These results indicate that explicit evidence organization can provide useful structured supervision for fact-checking.

Several limitations delimit the scope of these conclusions. Main benchmark results use two 8B verifier backbones, and the cross-verifier mechanism check covers only one 4B and one 8B model; most component and capacity ablations remain under Ministral. Broader conclusions about model families and scales, and confirmation of the observed ordering trend, require further evaluation. EviTrace relies on upstream candidate recall and LLM-generated structured annotations; although experiments show that LLM annotations possess a certain degree of reliability, annotation bias or recall failures may still lead to error propagation. On the other hand, this paper focuses solely on evidence selection and does not address how to retrieve and incorporate new supplementary evidence; to achieve optimal generality, it is necessary to incorporate a broader range of external evidence sources, such as those that support continuous online search. Meanwhile, this paper also does not explicitly handle scenarios in which evidence presents conflicting stances; conflicting evidence serves as a state-change signal in our method, while the spe-



Figure 4: An example evidence trace produced by EviTrace. Claim atoms, selected evidence units, atom-state transitions, and the final verifier output are shown.

cific handling is delegated to the LLM verifier, which makes this component less interpretable than certain defense-based methods. Finally, the methodological boundary of EviTrace lies in externalizing evidence reasoning to the selector level; it does not provide a causal explanation for the verifier’s latent reasoning or predictions. Our future work will incorporate more evidence sources, consider low-cost methods that do not require LLM API annotation, and construct more explicit conflict-resolution strategies.

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